

Advanced (Carolina Reaper)

Q1. Rationalize the denominator: $\frac{1}{\sqrt{2}+1}$

- (A) $\frac{1}{\sqrt{2}+1} \cdot \frac{\sqrt{2}-1}{\sqrt{2}-1}$
(B) $\frac{1}{\sqrt{2}+1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1}$
(C) $\frac{\sqrt{2}-1}{2-1}$
(D) $\frac{\sqrt{2}+1}{2+1}$
(E) $\frac{\sqrt{2}-1}{3}$

Q2. A furniture store is having a sale on desks. The original price of a desk is 200. If the sale price is $200\sqrt{2}$, what is the sale price after rationalizing the denominator?

- (A) $100\sqrt{2}$
(B) $100\sqrt{3}$
(C) 141.42
(D) $200\sqrt{2}$
(E) 282.84

Q3. Rationalize the denominator: $\frac{3}{\sqrt{5}-2}$

- (A) $\frac{3}{\sqrt{5}-2} \cdot \frac{\sqrt{5}+2}{\sqrt{5}+2}$
(B) $\frac{3}{\sqrt{5}-2} \cdot \frac{\sqrt{5}-2}{\sqrt{5}-2}$
(C) $\frac{3(\sqrt{5}+2)}{5-4}$
(D) $\frac{3(\sqrt{5}-2)}{5+4}$
(E) $\frac{3(\sqrt{5}+2)}{3}$

Q4. A surfer is riding a wave with a height of $5\sqrt{3}$ feet. If the wave's height is decreasing at a rate of $\frac{1}{\sqrt{3}}$ feet per second, what is the wave's height after 2 seconds?

- (A) $5\sqrt{3} - 2\sqrt{3}$
(B) $5\sqrt{3} - \frac{2}{\sqrt{3}}$
(C) $5\sqrt{3} - \frac{2\sqrt{3}}{3}$
(D) $5\sqrt{3} - 2$
(E) $5\sqrt{3} + 2$

Q5. Rationalize the denominator: $\frac{2}{\sqrt{7}+\sqrt{3}}$

- (A) $\frac{2}{\sqrt{7}+\sqrt{3}} \cdot \frac{\sqrt{7}-\sqrt{3}}{\sqrt{7}-\sqrt{3}}$
(B) $\frac{2}{\sqrt{7}+\sqrt{3}} \cdot \frac{\sqrt{7}+\sqrt{3}}{\sqrt{7}+\sqrt{3}}$
(C) $\frac{2(\sqrt{7}-\sqrt{3})}{7-3}$
(D) $\frac{2(\sqrt{7}+\sqrt{3})}{7+3}$
(E) $\frac{2(\sqrt{7}-\sqrt{3})}{4}$

Q6. A cyclist is traveling at a speed of $15\sqrt{2}$ miles per hour. If the cyclist is traveling for $\frac{1}{\sqrt{2}}$ hours, how far will the cyclist travel?

- (A) 15 miles
(B) $15\sqrt{2}$ miles
(C) 10.61 miles
(D) 21.21 miles
(E) 30 miles

Q7. Rationalize the denominator: $\frac{5}{\sqrt{11}-2}$

- (A) $\frac{5}{\sqrt{11}-2} \cdot \frac{\sqrt{11}+2}{\sqrt{11}+2}$
(B) $\frac{5}{\sqrt{11}-2} \cdot \frac{\sqrt{11}-2}{\sqrt{11}-2}$
(C) $\frac{5(\sqrt{11}+2)}{11-4}$
(D) $\frac{5(\sqrt{11}-2)}{11+4}$
(E) $\frac{5(\sqrt{11}+2)}{7}$

Q8. A water tank can hold $2000\sqrt{2}$ gallons of water. If $\frac{1}{\sqrt{2}}$ of the tank is already filled, how many gallons of water are already in the tank?

- (A) 1000 gallons
(B) 1414.21 gallons
(C) 2000 gallons
(D) 2828.43 gallons
(E) 4000 gallons

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify: $\frac{3\sqrt{6}+2}{\sqrt{2}+\sqrt{3}}$

Q10. A carpenter is building a staircase with a height of $10\sqrt{5}$ feet. If the staircase has $\frac{1}{\sqrt{5}}$ times the height of the building, what is the height of the building?

Chef's Tips

- Tip 1: Encourage students to check their work by plugging their answers back into the original equation.
- Tip 2: Remind students to rationalize the denominator by multiplying the numerator and denominator by the conjugate of the denominator.
- Tip 3: Suggest that students use a calculator to check their answers, but also require them to show their work and explain their reasoning.